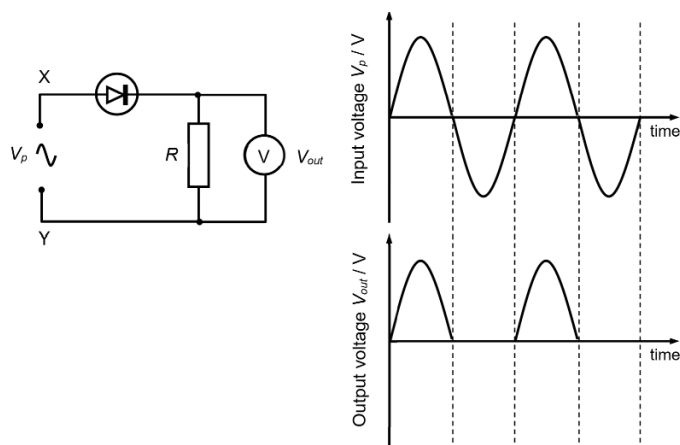


From s - t graph, the v - t graph is similar to the graph in C. Hence, answer is C.

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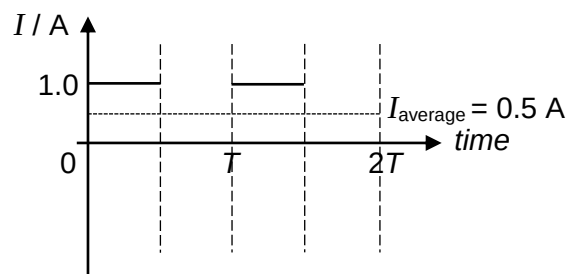
A

The input and output voltage due to single diode rectification is as shown below.

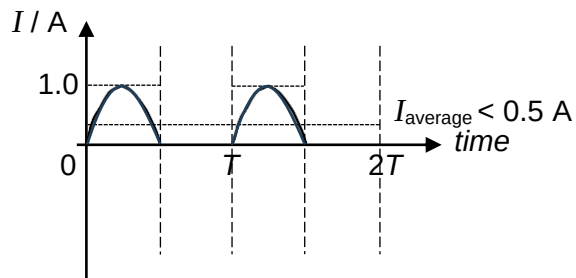


Since the peak voltage is 100 V and $R = 100 \Omega$, the peak current is 1.0 A.

Consider a current which varies with time in the following manner with respect to time. The average current would be 0.5 A as the 1.0 A current is averaged out equally over one period.



For a single diode rectified sinusoidal current, the mean will be less than 0.5 A as for most of the time the instantaneous current is less than 1.0 A as shown below.



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B